

Literature Review on Credit Market and Statistical Physics

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Abstract

This literature review focuses on the application of statistical physics to credit market analysis, with a special emphasis on structural credit risk models. It attempts to examine two pivotal studies and their interdisciplinary methodologies, providing proof that statistical physics provides a structural framework for financial modeling and analysis.

Contents

1	Introduction	2
2	Statistical Physics for Financial Modeling	2
2.1	Microstates and the Boltzmann Distribution	2
2.2	Entropy and the Maximum Entropy Principle	2
2.3	Statistical Ensembles	3
2.4	Brownian Motion	3
3	Application of Statistical Physics to Credit Markets	3
3.1	Ensemble Approaches to Credit Markets	4
3.2	Structural Credit Risk Modeling	5
3.3	Application	8
4	From the limitations of credit market modeling to complex systems	9
5	Conclusion	10
6	References	11

1 Introduction

The application of statistical physics to finance has become mandatory for a profound understanding of modern financial analysis [2]. Using the tools of equilibrium statistical mechanics, we can study the process for the empirical distribution of income and wealth in a closed system. Although this concept was criticized due to the impossibility of considering a closed system in finance, an attempt has been made by *Stefano Viaggiu et al.*[1] to build a formalism where credit is introduced as a new variable in the monetary base, allowing for a more realistic representation of the credit market. In [3, 4], the framework of structural credit risk is built, and then the authors focus on applying the analytical model to a homogeneous portfolio ¹.

The goal of this literature review is to introduce physicists to the application of statistical physics in the economy, particularly structural credit risk modeling. It seeks to highlight both the strengths and the ongoing challenges for this type of model.

2 Statistical Physics for Financial Modeling

This section follows [6].

2.1 Microstates and the Boltzmann Distribution

Consider a system that can exist in a large number of microstates, each characterized by an energy E_i . In thermal equilibrium, and setting the Boltzmann constant to unity, the probability that the system occupies a state with an energy ϵ is given by the Boltzmann distribution:

$$P(\epsilon) = \frac{1}{Z} e^{-\frac{\epsilon}{T}}, \quad (1)$$

where T is the temperature and Z is the partition function defined as:

$$Z = \sum_i e^{-\frac{\epsilon_i}{T}}. \quad (2)$$

Considering that the sum is taken over all states of the system, the expectation value of any physical variable x is given by

$$\langle x \rangle = \frac{\sum_k x_k e^{-\frac{\epsilon_k}{T}}}{\sum_k e^{-\frac{\epsilon_k}{T}}} \quad (3)$$

2.2 Entropy and the Maximum Entropy Principle

The entropy S of the system quantifies the degree of uncertainty or the number of microstates accessible to the system:

$$S = - \sum_i P_i \ln P_i. \quad (4)$$

By applying the maximum entropy principle, one obtains the equilibrium probability distribution by maximizing S subject to the constraints of normalization and a fixed average energy:

$$\begin{aligned} &\text{Maximize } S = - \sum_i P_i \ln P_i, \\ &\text{subject to } \sum_i P_i = 1 \quad \text{and} \quad \sum_i E_i P_i = \langle E \rangle. \end{aligned} \quad (5)$$

¹A portfolio is a compilation of financial assets, including stocks, bonds, cash, and other investments, owned by an individual or institution. Portfolios are structured to equilibrate risk and return in accordance with the investor's objectives, risk appetite, and investment horizon [5].

2.3 Statistical Ensembles

In the canonical ensemble, the system is in thermal equilibrium with a heat bath at temperature T . We can calculate the system's average energy as follows:

$$\langle E \rangle = \sum_i E_i P_i. \quad (6)$$

The free energy F is a key thermodynamic potential defined by:

$$F = -k_B T \ln Z. \quad (7)$$

This quantity connects the microscopic properties of the system with its macroscopic behavior and is instrumental in understanding phase transitions and response functions.

Alternatively, in the microcanonical ensemble, the system is isolated with a fixed total energy E . The entropy in this context is related to the number of accessible microstates $\Omega(E)$:

$$S = k_B \ln \Omega(E). \quad (8)$$

2.4 Brownian Motion

It refers to the random movement of microscopic particles suspended in a fluid due to incessant collisions with the much smaller, rapidly moving molecules of the medium [6, 7]. A common way to describe this behavior is through the Langevin equation, which for a particle of mass m reads

$$m \frac{dv}{dt} = -\gamma v + \xi(t), \quad (9)$$

where γ is the friction coefficient and $\xi(t)$ is a random force. The mean square displacement in one dimension is given by

$$\langle x^2(t) \rangle = 2Dt, \quad (10)$$

with the diffusion coefficient $D = \frac{k_B T}{\gamma}$. This result, which connects microscopic dynamics to macroscopic diffusion, is discussed in detail in [7]. A complementary approach involves the Fokker–Planck equation, which governs the time evolution of the probability density $P(x, t)$ for finding a particle at position x at time t . For Brownian motion, the Fokker–Planck equation is written as

$$\frac{\partial P(x, t)}{\partial t} = D \frac{\partial^2 P(x, t)}{\partial x^2}. \quad (11)$$

Given an initial condition $P(x, 0) = \delta(x)$, the solution is the Gaussian

$$P(x, t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right), \quad (12)$$

which describes how the probability distribution spreads out over time.

3 Application of Statistical Physics to Credit Markets

The framework provided by statistical physics led to modeling the complex dynamics of credit markets². This section is divided into the ensemble approaches to the credit market [1, 2], and the structural credit risk models discussed in [3, 4].

²A credit market is a financial arena for borrowing and lending activities. In these markets, diverse participants—including banks, firms, governments, and individuals—engage in credit transactions, wherein funds are provided immediately in exchange for future repayment with interest [5].

3.1 Ensemble Approaches to Credit Markets

Consider an isolated economic system with N agents³, in which the total money M remains constant. By integrating over the complete configuration space defined by the microscopic variables $\{x_i, y_i\}$ (where x_i denotes the cash held by the i th agent and y_i represents the credit or debt associated with that agent), and assuming the conservation of total money with $\overline{M} = m = \text{const}$, we introduce a thick shell Δ (with $\Delta \ll m$) to define the number of accessible microstates as

$$\Gamma(m) = \int_{m < M < m + \Delta} \frac{d^N x d^N y}{k^{2N}}, \quad (13)$$

where k is a normalization constant.

To facilitate the computation, we define the cumulative integral

$$\Sigma(m) = \int_{M \leq m} d^N x d^N y. \quad (14)$$

Then, the number of microstates within the shell Δ can be approximated by

$$\Gamma(m) = \Sigma(m + \Delta) - \Sigma(m) \approx \frac{\partial \Sigma(m)}{\partial m} \Delta. \quad (15)$$

Thus, the entropy of the system is given by

$$S = \ln \Gamma(m) = \ln \Sigma(m), \quad (16)$$

which effectively quantifies the logarithm of the number of microscopic configurations compatible with a fixed total money m for a large number of agents.

Model of a Credit Market

The ensemble formalism can be used to show a credit market, which, unlike a simple payment system, needs agents to follow certain accounting identities when they interact with each other. In a credit market, when one agent extends a loan of Δm to another, the accounting for each agent is maintained by the identities

$$M_1 = (x_1 - \Delta m) + (d_1 + \Delta m) = \text{Cte}_1, \quad (17)$$

$$M_2 = (x_2 + \Delta m) + (d_2 - \Delta m) = \text{Cte}_2, \quad (18)$$

where x_i denotes the cash held by agent i and d_i represents the net debt (i.e., assets minus liabilities) of agent i . These equations guarantee that the overall monetary base stays constant when agents exchange credit.

Additional constraints must be enforced on the system:

$$\sum_{i=1}^N x_i = M_0, \quad (19)$$

$$\sum_{i=1}^N (\text{assets})_i - \sum_{i=1}^N (\text{liabilities})_i = Q_0, \quad (20)$$

where M_0 is the fixed monetary base and, for simplicity, one may assume $Q_0 = 0$, resulting in similar distributions of credit and debt.

³In economics, a "agent" refers to any individual, firm, or entity that engages in decision-making and action-taking within the economic framework. These agents engage with one another, react to incentives, and affect market results by making decisions regarding consumption, production, investment, and various economic activities [5].

The credit market framework reinterprets the monetary function to exclusively focus on the credit variable.

$$\sum_{i=1}^N y_i = m, \quad \text{with } y_i \in [0, +\infty). \quad (21)$$

Since the monetary base M_0 is fixed, variations in the total money supply (within the canonical ensemble) arise solely from changes in credit. Thus, the partition function is expressed as

$$Z = \int d^N x d^N y \exp \left(-\frac{1}{T} \sum_{i=1}^N y_i \right). \quad (22)$$

Assuming that integration over the cash variables yields a factor proportional to M_0^N , the partition function factorizes as

$$Z = M_0^N T^N. \quad (23)$$

Therefore, the effective temperature of the credit market is defined as

$$T = \frac{m}{N}, \quad (24)$$

which represents the average credit exposure per agent. Introducing a generalization of the simple money functions, i.e. $M = \sum_{i=1}^N (x_i + y_i)$,

$$M = \sum_{i=1}^N \sum_{j=1}^I y_{ij} \quad (25)$$

where the index i runs over the agents and j over the different asset⁴ classes. Describing a full-fledged financial market can be obtained by

$$F = -TNI \ln T \quad (26)$$

$$S = NI \ln T + NI \quad (27)$$

$$T = \frac{m}{NI} \quad (28)$$

Consequently, permitting a greater variety of assets and liabilities within the system would diminish the probability of encountering substantial values of m_i and significant deviations from m , as T will be reduced for the same m . Diversification enhances system stability, aligning with conventional economic theory.

3.2 Structural Credit Risk Modeling

This section follows [3, 4]. Structural credit risk modeling quantifies credit risk by deriving loss distributions from the stochastic evolution of a firm's asset value. Traditional market risk measures rely on volatility, while credit risk modeling needs an asymmetric approach, i.e., losses only exist when default triggers [16, 17, 18, 19]. This structural framework begins with the asset price dynamics.

Let's start by considering a firm whose asset value $S(t)$ follows a geometric Brownian motion. But first, we need to define a single increment of a Brownian as

$$\Delta V_k(t) = \sigma_k (\sqrt{1 - c\epsilon_k} + \sqrt{c}\eta(t)) \quad (29)$$

⁴In economics, an asset is a valuable economic resource that can yield future advantages for its owner. Assets manifest in several forms, encompassing tangible goods like as real estate, machinery, or inventory, as well as intangible items including stocks, bonds, intellectual property, or goodwill [5].

where σ_k is the variance for asset k . For a single asset k , we can rewrite (29) as

$$(\Delta \vec{V}(t))_k = (\mathbf{A} \vec{B}(t))_k \quad (30)$$

The vector $\vec{B}(t)$ consists of the sum of all the assets k_i plus the correlation component $\eta(t)$. While the matrix $\mathbf{A}$ holds the correlation coefficients and variances. For instance, each are defined as follow

$$\mathbf{B}(t) = \begin{pmatrix} \epsilon_1(t) \\ \epsilon_2(t) \\ \vdots \\ \epsilon_K(t) \\ \eta(t) \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} \sigma_1 \sqrt{1-c} & 0 & \cdots & 0 & \sigma_1 \sqrt{c} \\ 0 & \sigma_2 \sqrt{1-c} & \cdots & 0 & \sigma_2 \sqrt{c} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \sigma_K \sqrt{1-c} & \sigma_K \sqrt{c} \end{pmatrix}.$$

To calculate the Brownian motion, we need to obtain the distribution of the asset prices for a certain time T

$$p(\vec{V}(T), \mathbf{A}, \mathbf{B}) = \int p(\mathbf{B}) \delta(\vec{V}(T) - \sum_{t=1}^T (\mathbf{A} \vec{B}(t)) d[\mathbf{B}]) \quad (31)$$

with $p(\mathbf{B})$ all probability densities of the random variables in $\mathbf{B}$. Using the Fourier transform, the delta distribution becomes

$$\delta(\vec{V}(T) - \sum_{t=1}^T (\mathbf{A} \vec{B}(t))) = \left(\frac{1}{2\pi}\right)^k \int \exp(-i\vec{\omega}^\dagger \vec{V}(T)) \exp(i\vec{\omega}^\dagger \sum_{t=1}^T \mathbf{A} \vec{B}(t)) d[\vec{\omega}] \quad (32)$$

Since modeling asset prices is a continuous-time stochastic process, we need to map the geometric Brownian motion to the ordinary Brownian motion [8]

$$V_k(T) \rightarrow \hat{V}_k(T) = \ln\left(\frac{V_k(T)}{V_{k,0}}\right) - \left(\mu_k - \frac{\sigma_k^2}{2}\right)T \quad (33)$$

Since we want to average over all possible correlation matrices and disclose the general statistical behavior of the system, we need to estimate the general impact of correlations. Thus, we would need to obtain the average price $\langle p(\vec{V}(T)) \rangle$ given by

$$\langle p(\vec{V}(T)) \rangle = \left(\frac{\sqrt{N}^K}{2\pi}\right) \int (-i\vec{\omega}^\dagger \vec{V}(T)) \frac{1}{(1 + (T/N)\vec{\omega}^\dagger \mathbf{S} \mathbf{S} \vec{\omega})^{N/2}} d[\vec{\omega}] \quad (34)$$

Using Gaussian integrals, Gamma function properties, and Bessel's function[3], (34) lead to

$$\langle p(\vec{V}(T)) \rangle = \left(\sqrt{\frac{N}{2\pi T}}\right)^K \frac{1}{\Gamma\left(\frac{N}{2}\right)} \left(\prod_{k=1}^K \frac{1}{\sigma_k V_k}\right) \times 2^{1-\frac{N}{2}} \left(\sqrt{\frac{N}{T} \sum_{k=1}^K \frac{\hat{V}_k^2}{\sigma_k^2}}\right)^{\frac{N-K}{2}} \kappa_{\frac{K-N}{2}} \left(\sqrt{\frac{N}{T} \sum_{k=1}^K \frac{\hat{V}_k^2}{\sigma_k^2}}\right) \quad (35)$$

where

$$\hat{V}_k(T) = \ln\left(\frac{V_k(T)}{V_{k,0}}\right) - \left(\mu_k - \frac{\sigma_k^2}{2}\right)T \quad (36)$$

the standard deviation is

$$\hat{\sigma}_k = \sqrt{\exp(2\mu + \sigma_k^2 T) (\exp(\sigma_k^2 T) - 1) V_{k,0}^2} \quad (37)$$

If we define a threshold value Δ_K , where $V_k(T) > \Delta_K$, then the model presented in [3] remains correct and produces correct results. But when a default occurs, namely the obligor's

equity⁵, the model needs an improvement by adding the loss distribution L_k to the mathematical framework. Now, let's denote the threshold as F_k , i.e., *the face value*. The distribution is defined as,

$$L_k = \begin{cases} \frac{F_k - V_k(T)}{F_k}, & \text{if } V_k(T) < F_k \quad (\text{default}), \\ 0, & \text{else (no default)}, \end{cases} \quad (38)$$

hence, the overall loss of a portfolio can be defined only when we weight each loss by its face value, which will be written as a sum

$$L = \sum_{k=1}^K f_k L_k, \quad f_k = \frac{F_k}{\sum_{l=1}^K F_l} \quad (39)$$

let's consider the following example: we have two credits ($K = 2$) in our portfolio

1. Credit 1: face value $F_1 = 50,000USD$, with a loss of 0.4
2. Credit 2: face value $F_2 = 90,000USD$, with a loss of 0.15

$$f_1 = \frac{F_1}{F_1 + F_2} = \frac{50,000}{50,000 + 90,000} = 0.35, \quad f_2 = \frac{90,000}{50,000 + 90,000} = 0.64. \quad (40)$$

Thus, the overall portfolio loss L is:

$$L = f_1 L_1 + f_2 L_2 = (0.35)(0.4) + (0.64)(0.15) = 0.236 \quad (41)$$

which corresponds to a 23.6% loss across the entire portfolio. Since the loss must be weighted by each credit's size compared to the total face value, and since we want to simulate real-world portfolios, we need to average the loss distribution [3, 9]. Assuming random correlations and an average correlation level of zero, the average loss distribution is given by

$$\langle p(L) \rangle \approx \frac{1}{\sqrt{2\pi} \Gamma(\frac{N}{2})} \int_0^\infty dz z^{\frac{N}{2}-1} e^{-z} \frac{1}{\sqrt{\widehat{M}_2(z)}} \exp\left(-\frac{(L - \widehat{M}_1(z))^2}{2 \widehat{M}_2(z)}\right) \quad (42)$$

with

$$\widehat{M}_1(z) = \sum_{k=1}^K f_k M_{1,k}(z), \quad \widehat{M}_2(z) = \sum_{k=1}^K f_k^2 (M_{2,k}(z) - M_{1,k}(z)^2) \quad (43)$$

Now with all the tools built in [3] and [4], we focus on the example of the homogeneous portfolio. A homogeneous portfolio is a portfolio where every credit k has the same face value $F_k = F$, the same asset value distribution parameters, and identical threshold conditions. With these assumptions, the weighting factor for every credit can be written

$$f_k = \frac{F_k}{\sum_{l=1}^K F_l} = \frac{1}{K} \quad (44)$$

then, the portfolio can be considered as an ensemble of K identical copies of a unique credit sample, i.e. the loss distribution L_k has the same distribution, which can be defined as

$$L = \sum_{k=1}^K \frac{1}{K} L_k \quad (45)$$

⁵The obligor's equity is the remaining worth of a firm's assets after the deduction of all liabilities. In credit risk and financial modeling, it denotes the net value or the equity stake of the firm's shareholders. In the Merton model—a structural framework for credit risk—equity is perceived as a call option on the firm's assets, retaining value alone if the assets surpass the debt obligations at maturity. When a firm's asset value declines beneath its liabilities, the obligor's equity becomes zero or becomes negative, frequently signifying default or bankruptcy [5].

mfuctions $\hat{M}_1(z)$ and $\hat{M}_2(z)$ become independent of k and can be written

$$\widehat{M}_1(z) = \sum_{k=1}^K f_k M_{1,k}(z) \rightarrow M_1(z), \quad \widehat{M}_2(z) = \sum_{k=1}^K f_k^2 (M_{2,k}(z) - M_{1,k}(z)^2) \rightarrow \frac{1}{K} (M_2(z) - M_1^2(z)) \quad (46)$$

Therefore, the average loss distribution will be written

$$\begin{aligned} \langle p(L) \rangle \approx & \frac{1}{2\pi T \Gamma(\frac{N}{2})} \sum_{j=0}^K \int_0^\infty dz z^{\frac{N}{2}-1} \exp(-z) \sqrt{\frac{2\pi K^2}{j(M_2(z) - M_1(z)^2)}} \\ & \times \exp\left[-\frac{(LK - jM_1(z))^2}{2j(M_2(z) - M_1(z)^2)}\right] \left(\frac{1}{2} + \frac{1}{2} \operatorname{erf}\left[\frac{\ln(F/V_0) - (\mu - \frac{\sigma^2}{2})T}{\sigma\sqrt{2T}}\right]\right)^{K-j}. \end{aligned} \quad (47)$$

3.3 Application

With the analytical model developed and the homogeneous portfolio defined, we can apply the analytical model to a word-like model. In [3, 4], several examples have been deployed; we focus on the second example where $K = 50$. The set of input parameters V_0, μ, σ, F , and T are defined using the table 1

Variable	Description	Value(s)	Unit
K	Number of assets	50	[–]
T	Time of maturity	1	[year]
σ_k	Volatility of the k -th asset	0.15	[year ^{-1/2}]
μ_k	Drift of the k -th asset	0.05	[year ⁻¹]
N	Parameter to control correlations	$\begin{cases} N=K \\ N \rightarrow \infty \end{cases}$	[–]
$V_{k,0}$	Start price of the k -th asset	100	[currency]
F_k	Face value of the k -th asset	75	[currency]

Table 1: Input Parameters for the simulation

The parameters above are needed to analyze the impact of correlations, i.e., the loss distribution for a homogeneous portfolio. Since we have the freedom to choose the input parameter N , we choose two numbers

1. the strongest impact of random correlations, $N = K$.
2. the uncorrelated limit, i.e., the correlation matrix becomes the identity matrix, $N \rightarrow \infty$

The result of the simulation is heavy tails of the loss distribution of the correlated portfolio compared to the uncorrelated case; this still true even when the distribution becomes narrower for large values of K . See 1. We can conclude that if the correlations are not modeled properly, the tails of the loss distribution are strongly underestimated. In this analysis, in order to simplify the model the normalization of the loss distribution is not exact, the maturity time ⁶

⁶Maturity time refers to the duration from the present until a financial instrument—such as a bond, loan, or derivative contract—attains its expiration date. Upon maturity, the principle or face value is to be repaid, and any outstanding contractual obligations, such as final interest payments, must be satisfied. This parameter is essential in finance since it affects present value computations, risk evaluations, and the total pricing of financial products [5].

T is a constant, and the drifts ⁷ μ_k and standard deviations σ_k are not set correctly to evaluate the evolution of the loss distribution.

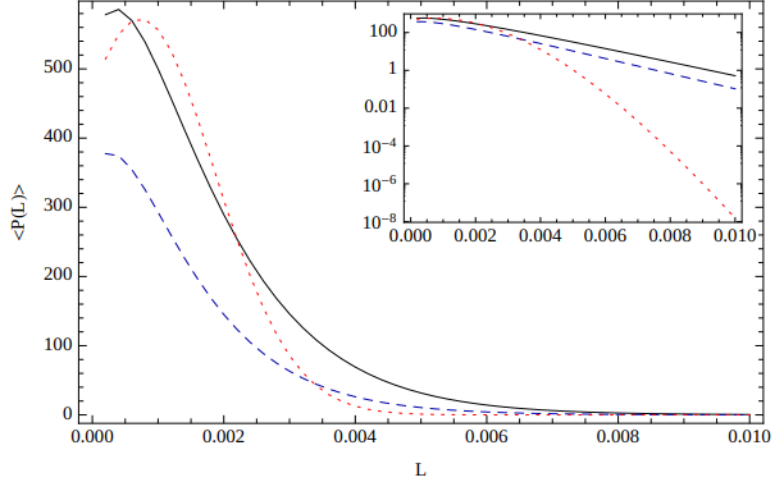


Figure 1: The loss distribution of a homogeneous portfolio with $\sigma = 0.15$, $\mu = 0.05$, $T = 1$, $V_0 = 100$, $F = 75$, and $K = 50$. The dashed line represents the simple approximation, and the solid line represents the improved approximation, both calculated with maximum random correlations, $N = K$. The dotted line represents the uncorrelated case, calculated by the approximation $N = 30K$.

4 From the limitations of credit market modeling to complex systems

One of the many lessons of the financial crisis of 2008-2009 was the improper estimation of credit risk, where used models underestimate risks embedded in credits. Assessing the risk of a credit portfolio cannot be done without considering correlations between obligors, in particular the treatment of debtors-in-possession in bankruptcy proceedings [3, 4]. Thus, even considering real-world parameters, current credit risk modeling often relies on assumptions of rationality in decision-making. Financial data consistently challenges these assumptions. Recognizing the complexities of human behavior and the limitations of current models is an urgent need for financial analysis [10].

The concept of *reflexivity* in credit markets needs to be introduced and may be the next main research area in econophysics and may open new areas in complex systems. These new models should be built on [11, 12, 13]:

1. credit markets are strongly correlated with investors' beliefs and market outcomes.
2. sentiments, positive or negative, can influence borrowing conditions and strengthen the current sentiment. This is particularly the case during booms and panic.
3. credit market expectations can lead to a strong divergence from economic fundamentals output.
4. group behavior, acting like a larger group in economy, can impact the stability of the credit market dynamic.

⁷In economic and financial models, "drift" refers to the average or expected change of a variable over time. [5]

The 4th point opens other doors to studying the stability of the economy. We may consider in the future applying phase transitions and criticality to understand transitions between stability and instability in finance and investigate equilibrium and out-of-equilibrium behavior in credit markets, which cannot be done without adapting techniques like the method of critical fluctuation [14, 15].

5 Conclusion

This review discusses the application of statistical physics to credit market analysis, focusing on structural credit risk modeling. The literature [1, 2, 3, 4] proved the strength of econophysics by providing the correct analytical models to build real-like world applications.

Although statistical physics provides a structural framework for financial modeling and analysis, and despite the promising developments, significant challenges remain. Current models are unable to capture the non-stationary nature of real-world credit markets. This limit highlights the urgency for future research to incorporate the dynamic and evolution of real credit markets into econophysics [10].

Concepts such as reflexivity and group behavior [11, 12, 13], using techniques from complex systems like phase transitions and criticality, could really provide usable and proven models for financial analysis; this would ultimately lead to better economic growth, providing accurate and correct, driven-decision results [14, 15].

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